FiberCANDI - Deconvolving Bulk Assays to Single Molecule Resolution

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1 Research problem

The problem setup. We study a weakly supervised regression problem from genomics. In weak supervision, labels are coarser than the predictions we want: they come at the level of groups of instances, not the instances themselves. Even through this resolution mismatch, we want to learn a model that predicts at the instance level. Each example is a *bag* of up to 100 sequences (100 instances), one per DNA molecule, aligned to the same window of a 1-D coordinate axis over the genome. The label is a real-valued signal over that window, but it describes only the *average* over a huge population of such molecules, never a single one. We want a model whose *per-molecule* outputs are correct. **Main Question:** can per-instance predictions be learned from bag-average labels alone, and can they be verified when no instance-level labels exist? Below we describe the data formats, the biological context, and a falsifiable evaluation plan. Further details on the biology are in Appendix A. The data is summarized in Table 1.

Stage 1: coordinates and the reference. The human genome is a string of 3×10^9 letters over $\{A, C, G, T\}$; a position x is an index into it. We work on windows given as (*chromosome, start, end*), with $L = \text{end} - \text{start} = 5{,}000$ positions. The window's reference string is given to the model one-hot, $S \in \{0, 1\}^{4\times L}$. It is the same in every cell and molecule, so it tells the model *where* it is, but not *which* cell type it is looking at.

Stage 2: reads, the instances. In a cell, DNA is wound around protein spools (*nucleosomes*); the stretches between spools are *accessible* to the cell's machinery. Which stretches are open differs between cell types, and between individual molecules of one cell type. The Fiber-seq assay chemically marks the open bases of every molecule, then reads each molecule (a *fiber*, here 5–25 kb long) end to end [Stergachis et al. 2020]. After standard processing [Jha et al. 2024], a read becomes $C = 5$ event channels along the coordinate axis: marked base (an A), methylated CpG (a second chemical mark, at C-G letter pairs), open patch, nucleosome, and strongly open regulatory element. Each channel takes +1 (event present), −1 (measurable and absent) or 0 (not measurable: outside the read, or a letter where the event cannot occur). The $N \le 100$ reads overlapping a window form a tensor of *instances* $X \in \{-1, 0, 1\}^{C\times L\times N}$, with a coverage mask $M \in \{0, 1\}^{L\times N}$. The reads are an unordered set.

Stage 3: targets, the bag labels. We want $K = 8$ other properties for every read; the experiments that measure such properties are called *assays*. Six are chemical tags on the nucleosomes (*histone marks*, e.g. H3K27ac) that indicate whether nearby genes are on or off. The other two are binding of the protein CTCF and accessibility as measured by a second assay (ATAC-seq). No technology measures these at single-molecule resolution. They are measured in *bulk*: an assay pools millions of cells of one cell type and reports, per position, how enriched the property is relative to a control (*fold change*, FC). The result is a track $y_k \in \mathbb{R}^L_{\ge 0}$ per property. ENCODE [ENCODE Project Consortium 2020] provides these tracks for our 8 cell types, which are cultured cell lines such as GM12878 (blood) and HepG2 (liver). Since all cell types share one genome, every difference between their tracks has to come from cell-specific factors.

Stage 4: per-molecule output and aggregation. The model maps each read, with the reference as context, to K per-position values $z_f \in \mathbb{R}^{K\times L}_{\ge 0}$; these are the imputation (predictions). Reads are processed independently, so no information passes between instances, and the stacked output $Z \in \mathbb{R}^{K\times L\times N}_{\ge 0}$ is a set of per-molecule predictions. The label exists only at bag level, so the outputs are averaged over the reads covering each position and compared with the track. Values are kept in asinh space, which compresses the signal's range and makes it easier to learn, but the average is taken in FC space, where bulk signal adds up over molecules:

$$\hat{y}_k(x) = \operatorname{asinh}\left(\tfrac{1}{|F_x|}\textstyle\sum_{f\in F_x} \sinh z_{f,k}(x)\right), \tag{1}$$

$$\mathcal{L} = \textstyle\sum_{k,x} w_{k,x}\left(\hat{y}_k(x) - \operatorname{asinh} y_k(x)\right)^2. \tag{2}$$

Here $F_x = \{f : M_{x,f} = 1\}$ is the set of reads covering x, and the weight w keeps only positions covered by at least 5 reads, to avoid noisy averages. Training therefore sees a 2-D label ($K \times L$), while the deliverable is the 3-D per-molecule tensor Z.

Table 1. The data for one window (L = 5,000 positions, x the genomic position).

Object	Shape	Values	Role
Reference S	$4 \times L$	one-hot	input, shared by cell types
Reads X	$C \times L \times N$	$\{-1, 0, 1\}$	input, $C = 5$, $N \le 100$
Coverage M	$L \times N$	$\{0, 1\}$	which reads cover x
Output Z	$K \times L \times N$	$\mathbb{R}_{\ge 0}$	per-molecule imputation
Aggregate $\hat{Y}$	$K \times L$	$\mathbb{R}_{\ge 0}$	bag prediction (Eq. 1)
Track Y	$K \times L$	$\mathbb{R}_{\ge 0}$	bag label (bulk)

Why it matters. Bulk tracks hide what happens on individual molecules: differences between the two parental copies of a gene, sub-populations of cells, and which properties co-occur on the same molecule. Fiber-seq already reads millions of molecules. Imputing the bulk properties onto each of them would turn existing Fiber-seq data into a single-molecule atlas of many properties at once.

Assumptions. (A1) A molecule's accessibility and nucleosome pattern, together with its sequence, carry information about its marks. (A2) A FC track is linear in the fraction of marked molecules,

up to a per-experiment scale. (A3) The Fiber-seq and ENCODE samples of a cell line describe the same population.

Evidence plan. Our hypothesis (H) is that the per-fiber outputs carry information specific to each molecule. A model can also fit bulk tracks by predicting the bulk on every molecule (our null, H0), so bulk accuracy is necessary but not sufficient. Throughout, p is a predicted bulk track (the average of a cell type's fibers, $\hat{y}_k$ in Eq. 1), t is that cell type's ENCODE track ($\operatorname{asinh} y_k$), and all metrics pool every scored base of the test chromosome. We test H against H0 at three levels.

(i) *Bulk recovery* on held-out chromosomes and cell types. We use 5 development and 3 held-out cell lines; the held-out ones range from near to far from the development set. We report the Pearson correlation $r_{pt} \in [-1, 1]$ (1: the predicted shape along the genome matches the target up to scale and offset; 0: no linear relation; -1: inverted) and the mean squared error between p and t (≥ 0, in squared asinh units; 0 is a perfect prediction). As a baseline we use the *cross-cell average* b: at each position, the mean of the training cell lines' tracks, which needs no information about the evaluated cell type. Much of every track is shared across cell types, so b is a strong baseline. To isolate what is specific to the cell type, we report the *cell-specific correlation*, the partial correlation of p and t given b. Both are regressed on b, and their residuals are correlated:

$$r(p, t \mid b) = \operatorname{corr}\left(p - \hat{p}(b),\ t - \hat{t}(b)\right).$$

A simpler way to compute the same quantity exactly is

$$r(p, t \mid b) = \frac{r_{pt} - r_{bt} r_{pb}}{\sqrt{(1 - r_{bt}^2)(1 - r_{pb}^2)}}.$$

It lies in $[-1, 1]$: 1 means the model predicts exactly where and how this cell type departs from the average; 0 means nothing beyond the average (the baseline itself, or any rescaling of it, scores exactly 0); negative values mean the predicted departures point the wrong way.

(ii) *Mixture-only supervision* (main test). We train only on bags that mix reads from two cell types A and B in varying proportions, labelled with the mixture's track. Neither cell type's own track is ever used, so the model never sees a label for either population on its own. We then run each cell type's fibers alone, giving p_A and p_B, and compare them with t_A and t_B. Under H0 the model predicts the mixture for both, so $p_A \approx p_B$. Each of the following metrics has a known value under H0:

- *Differential correlation* $\operatorname{corr}(p_A - p_B,\ t_A - t_B)$, per base over positions scored in both cell types: does the predicted difference between the two cell types follow the true one? Range $[-1, 1]$: 1 means the predicted differences follow the true ones exactly (up to scale), 0 means no relation (value for H0), and -1 means they are systematically reversed.
- *Sign accuracy*: over *differential bins* (1 kb bins where $|t_A - t_B| \geq 0.5$), the fraction where the mean of $p_A - p_B$ over the bin has the same sign as $t_A - t_B$. Range $[0, 1]$: 1 means the higher cell type is identified at every bin, 0.5 is chance (value under H0), and 0 means it is always reversed.
- *Fiber-origin AUC*: at each differential bin, score each fiber by its mean output z_f over the bin. The AUC is the probability that a random fiber from the cell type with the higher track scores above a random fiber from the other; we average it over bins. It asks whether a *single molecule's* prediction reveals where it came from. Range $[0, 1]$: 1 means every fiber of the cell type with the mark outranks every fiber of the other (perfect separation), 0.5 means no separation (value under H0), and 0 means perfect separation in the wrong direction. Even a perfect model stays below 1, because not every molecule of a cell type carries its mark.

As a control for library effects, we use two Fiber-seq libraries of one cell line, which share one true track: any separation between them comes from library differences, not biology.

(iii) *Within-bag structure.* We use properties known to differ between molecules of one cell type. Examples are an input channel held out and checked against the outputs (CpG methylation), and the X chromosome in female cells, where one of the two copies is switched off. The outputs must vary in the same way.

Preliminary results. (i) Over all 6,843 test tiles of chr21, the cell-specific correlation is positive in all 8 cell lines (+0.19 to +0.41; +0.19 to +0.27 in cell lines never seen in training). (ii) In the mixture test (GM12878 + HepG2), the model trained only on mixtures separates the two populations' molecules as well as a model trained on each cell type's own track (Table 2).

Table 2. Mixture test, mean over 8 assays.

	H0	Mixture only	Own tracks
Differential r (5 kb windows)	0	0.49	0.47
Sign accuracy (differential bins)	0.50	0.78	0.74
Fiber-origin AUC	0.50	0.64	0.60

2 Literature review

Single-molecule chromatin profiling. Fiber-seq [Stergachis et al. 2020] marks accessible DNA with a methyltransferase and reads it with long reads, giving accessibility for every molecule. Single-cell CUT&Tag [Bartosovic et al. 2021] profiles marks in single cells, but with sparse coverage and only one or a few marks per cell. Together, these show that *measuring* many marks on one molecule is still out of reach. We instead *impute* them from Fiber-seq reads.

Epigenome imputation. ChromImpute [Ernst and Kellis 2015], PREDICTD [Durham et al. 2018] and Avocado [Schreiber et al. 2020a] predict missing bulk tracks from other tracks of the same cell type. They use regression or tensor factorization over cell types × assays × positions. CANDI [Foroozandeh et al. 2025] imputes and denoises bulk assays with self-supervision and calibrated confidence. dHICA [Wen et al. 2024] shows that a single input assay can be enough: it predicts several histone-mark tracks from bulk chromatin accessibility (ATAC-seq or DNase-seq) and DNA sequence alone, the bulk analogue of our input. All of them output population averages and need bulk tracks of the target cell type. FiberCANDI keeps the target (ENCODE assays) but changes the unit of prediction from a track to a molecule: its input is accessibility measured on each molecule, so it needs only the target cell type's fibers. Schreiber et al. [2020b] showed that models predicting across cell types often fail to

beat the average across training cell types. We adopt that baseline and score the partial correlation given it.

Sequence-to-function models. Enformer [Avsec et al. 2021] predicts thousands of tracks from DNA alone, using long-range attention. Its cell-type specificity comes from one output head per training cell type, so it cannot describe a new cell type or individual molecules. In our model the DNA is shared context, and the fibers carry everything specific to the cell and the molecule.

Weak supervision. Learning from label proportions [Quadrianto et al. 2009] recovers labels of individual instances from bag-level aggregates, which is possible only when bags differ in composition. Our setting is a regression analogue: a bag is a fiber stack and its label is the bulk track. This is why the mixture test varies the proportions: varying them makes each population's signal identifiable.

Compared with these works, we connect bulk-assay imputation (the second theme) to single-molecule inputs (the first), and we give a falsifiable verification protocol for a setting with no molecule-level labels.

3 Venue, contributions, and bibliography

Venue. MLCB 2027 (Machine Learning in Computational Biology). Its focus on new machine-learning formulations with biological evaluation matches our contribution: a weakly supervised model and a falsifiable test of what the model learns.

Prior work. Two lines of prior work underlie this project. Bulk-assay imputation with CANDI [Foroozandeh et al. 2025] defines the assay-imputation setting we extend to single molecules. Fiber-seq [Stergachis et al. 2020] provides the single-molecule chromatin data and its processing [Jha et al. 2024]. Bulk targets and cCRE annotations come from ENCODE [ENCODE Project Consortium 2020].

Contributions. The work in this proposal is mine (Azaan Rehman):

(1) The problem formulation, and a reproducible data pipeline.
(2) Fiber aggregation consistent with fold change (Eq. 1). It gives each per-molecule output a physical meaning and makes mixtures exact.
(3) An evaluation framework that separates bulk recovery from single-molecule information: the cross-cell baseline with partial correlation, cell-specific and pairwise loci, and the mixture-only test with a library control.
(4) First evidence that the per-molecule imputations carry molecule-specific signal.

To our knowledge, this is the first method to impute several histone marks per single molecule. Unlike CANDI and earlier imputation methods, its output is one track per molecule rather than one per cell type.

References

Žiga Avsec, Vikram Agarwal, Daniel Visentin, et al. 2021. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods* 18, 10 (Oct. 2021), 1196–1203. doi:10.1038/s41592-021-01252-x

Marek Bartosovic, Mukund Kabbe, and Gonçalo Castelo-Branco. 2021. Single-cell CUT&Tag profiles histone modifications and transcription factors in complex tissues. *Nature Biotechnology* 39, 7 (April 2021), 825–835. doi:10.1038/s41587-021-00869-9

Timothy J. Durham, Maxwell W. Libbrecht, J. Jeffry Howbert, Jeff Bilmes, and William Stafford Noble. 2018. PREDICTD: PaRallel Epigenomics Data Imputation with Cloud-based Tensor Decomposition. *Nature Communications* 9, Article 1402 (April 2018). doi:10.1038/s41467-018-03635-9

ENCODE Project Consortium. 2020. Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature* 583, 7818 (July 2020), 699–710. doi:10.1038/s41586-020-2493-4

Jason Ernst and Manolis Kellis. 2015. Large-scale imputation of epigenomic datasets for systematic annotation of diverse human tissues. *Nature Biotechnology* 33, 4 (2015), 364–376. doi:10.1038/nbt.3157

Mehdi Foroozandeh et al. 2025. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* (2025). doi:10.1101/2025.01.23.634626 Preprint.

Anupam K. Jha, Stephanie C. Bohaczuk, Yizi Mao, et al. 2024. DNA-m6A calling and integrated long-read epigenetic and genetic analysis with fibertools. *Genome Research* 34, 11 (2024), 1976–1986. doi:10.1101/gr.279095.124

Novi Quadrianto, Alex J. Smola, Tibério S. Caetano, and Quoc V. Le. 2009. Estimating labels from label proportions. *Journal of Machine Learning Research* 10 (2009), 2349–2374. https://jmlr.org/papers/v10/quadrianto09a.html

Jacob Schreiber, Timothy Durham, Jeffrey Bilmes, and William Stafford Noble. 2020a. Avocado: a multi-scale deep tensor factorization method learns a latent representation of the human epigenome. *Genome Biology* 21, Article 81 (2020). doi:10.1186/s13059-020-01977-6

Jacob Schreiber, Ritambhara Singh, Jeffrey Bilmes, and William Stafford Noble. 2020b. A pitfall for machine learning methods aiming to predict across cell types. *Genome Biology* 21, Article 282 (2020). doi:10.1186/s13059-020-02177-y

Andrew B. Stergachis, Brian M. Debo, Eric Haugen, L. Stirling Churchman, and John A. Stamatoyannopoulos. 2020. Single-molecule regulatory architectures captured by chromatin fiber sequencing. *Science* 368, 6498 (June 2020), 1449–1454. doi:10.1126/science.aaz1646

Wen Wen, Jiaxin Zhong, Zhaoxi Zhang, Lijuan Jia, Tinyi Chu, Nating Wang, Charles G. Danko, and Zhong Wang. 2024. dHICA: a deep transformer-based model enables accurate histone imputation from chromatin accessibility. *Briefings in Bioinformatics* 25, 6, Article bbae459 (2024). doi:10.1093/bib/bbae459

A Biology background

- *Chromatin and nucleosomes.* In a cell, DNA is wrapped around protein spools called *nucleosomes*, about 147 bases each, like beads on a string. The DNA plus its proteins is *chromatin*. A *fiber* is one such DNA molecule, which in our case is a few thousand to tens of thousands of bases long.
- *Accessibility.* Stretches of DNA between nucleosomes are *accessible*: proteins can reach them. Regulatory elements (switches that turn genes on or off) are usually accessible in the cells where they are active. ATAC-seq measures accessibility in bulk.
- *Histone marks.* Nucleosomes carry chemical tags (e.g. H3K27ac, H3K4me3, H3K27me3), named after the protein (histone H3), the position (lysine 27) and the tag (acetyl, methyl). Different marks indicate active promoters, active enhancers, transcribed genes or silenced regions. *CTCF* is a DNA-binding protein that organizes the genome into loops.
- *Bulk assays.* ChIP-seq and ATAC-seq pool DNA from millions of cells and report, at each position, how enriched a mark is relative to a control (*fold change*). The result is a *track*: one number per base, averaged over all molecules. ENCODE provides such tracks for many cell lines (cultured cell populations such as GM12878, a B-cell line, and HepG2, a liver line).
- *Fiber-seq.* An enzyme adds a chemical tag (m6A) to accessible bases. Long-read sequencing then reads each molecule end to end, along with its tags, so accessibility, nucleosome positions and DNA methylation (at CpG sites) are known for *every individual molecule*, not just on average. The five input channels are: m6A-marked bases, methylated CpGs, accessible patches (MSPs), nucleosomes, and FIRE elements (accessible patches classified as regulatory elements).